AMERICAN UNIVERSITY OF BEIRUT
MAROUN SEMAAN FACULTY OF ENGINEERING & ARCHITECTURE

Department of Industrial Engineering and Management (IEM)

# Present Worth and Related Economic Criteria

Engineering Economy

# Comparing and Making Decisions

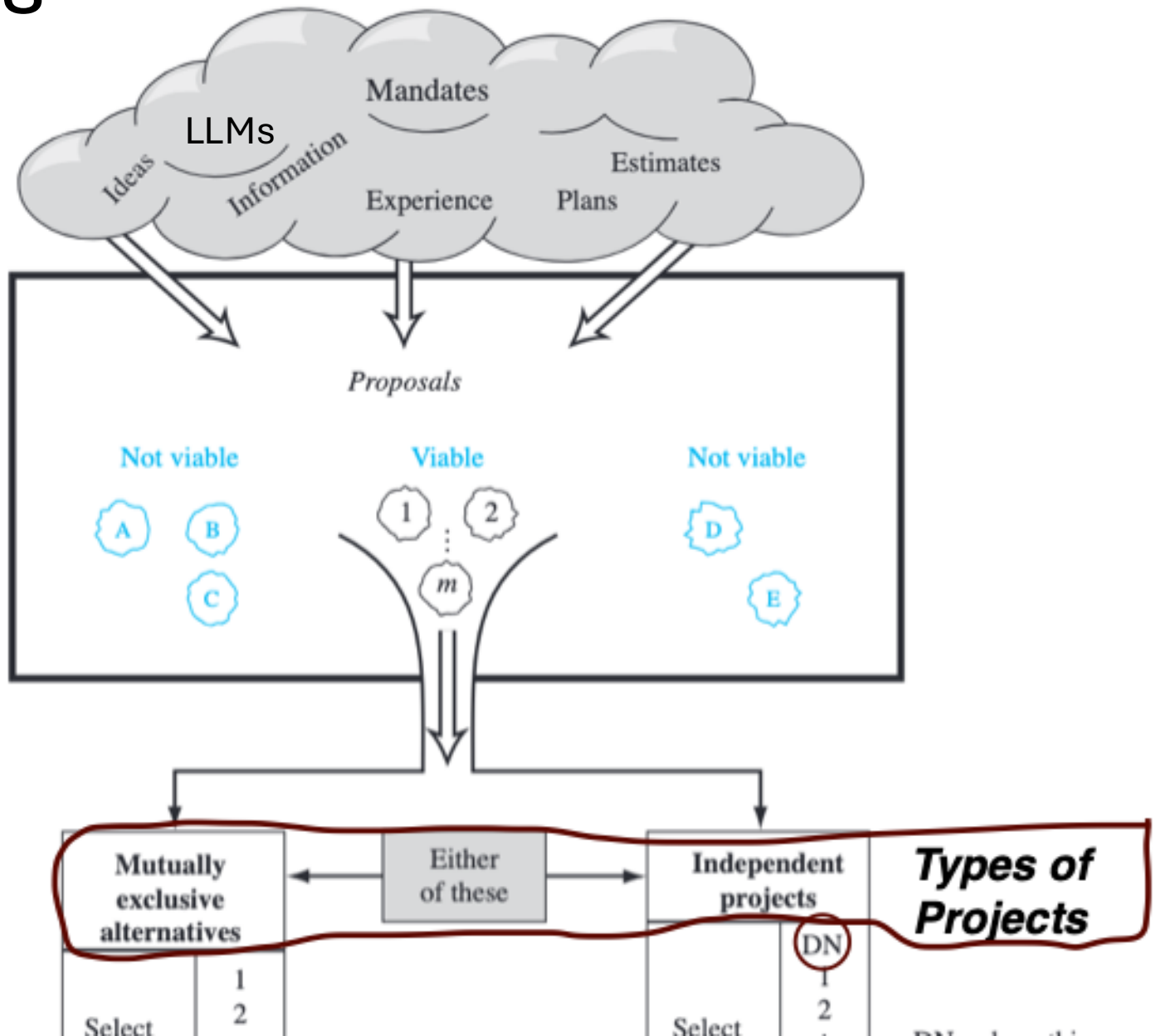


# Example – Upgrading a Production Facility

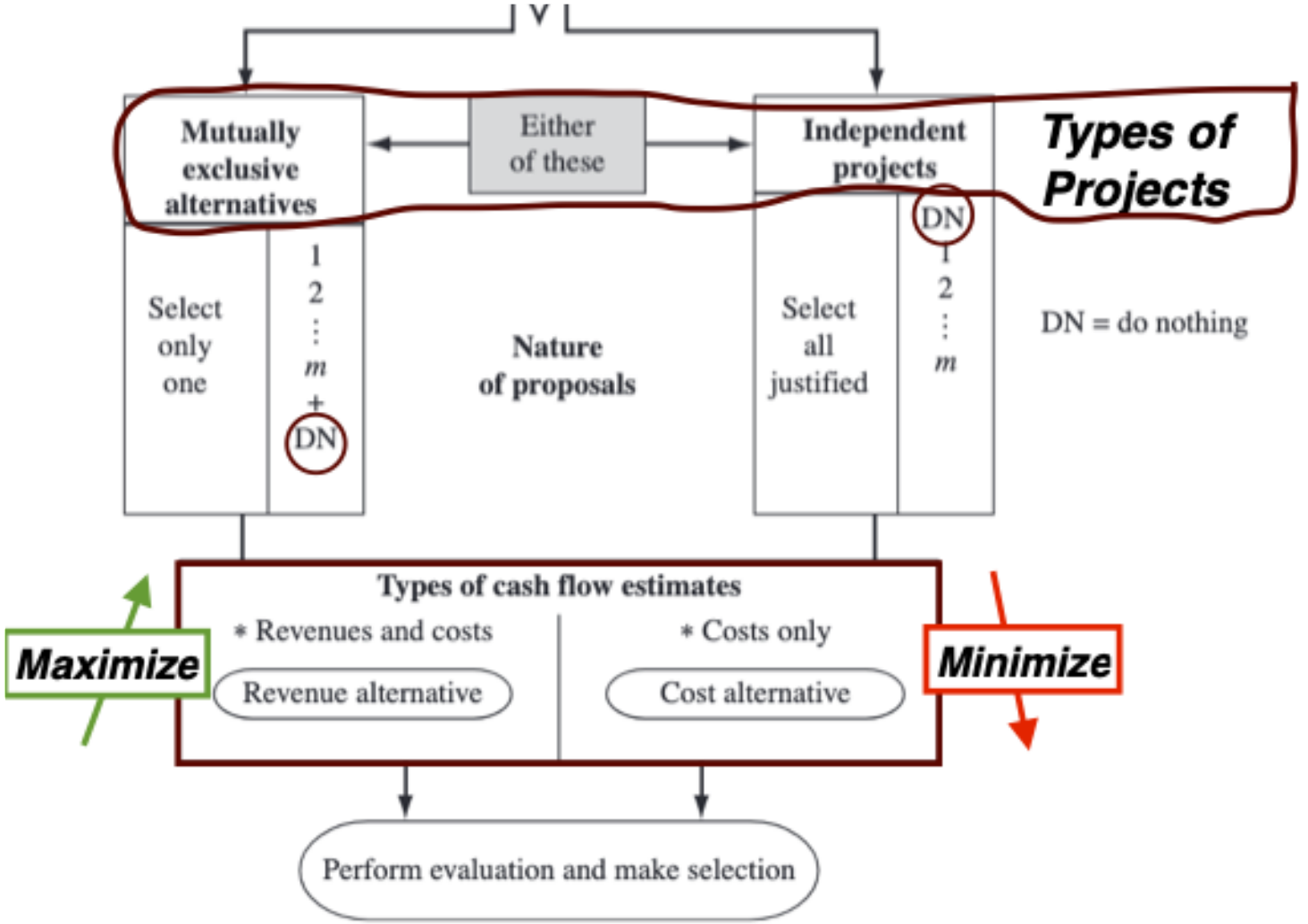
- A baby diaper production facility has one production and packaging line producing 200,000 diapers per day with 95% efficiency.

| Alt. | Description | Costs | Revenue |
|---|---|---|---|
| A | A new line with 240,000 throughput and 98% eff | 6 millions USD | 5,000 USD per day |
| B | A used line with 200,000 throughput and 96% eff | 2.5 millions USD | 2,000 USD per day |
| C | A *co-developed* line with 240,000 throughput and 97% eff | 5 millions USD | 3,000 USD per day |
| D | Install an online inspection system. | 100,000 Initial + 2,000 per year | 15,000 USD per year |
| E | Replace existing guiding systems with more accurate ones. | ± 16,000 Initial | 5,000 USD per year |
| F | Buy a new Cutting die made of tungsten carbide material. | 300,000 USD | 40,000 USD per year |

# Alternatives



# Alternatives



# Alternatives

| | **Mutually Exclusive**<br>(Projects compete; choose one) | **Independent**<br>(Projects stand alone; choose any) |
|---|---|---|
| **Cost Alternatives**<br><br>(Outflows only) | **Gas vs. Electric Truck**<br><br>• Need only one<br><br>• Neither generates income | **Fix Roof & Buy Forklift**<br><br>• Can fund both<br><br>• Neither generates income |
| **Revenue Alternatives**<br><br>(Inflows & Outflows) | **Hotel vs. Apartment**<br><br>• Same plot of land (pick one)<br><br>• Both generate income | **Stores in Beirut & Byblos**<br><br>• Can open both<br><br>• Both generate income |

# Example – Upgrading a Production Facility

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# Example – Upgrading a Production Facility

- A baby diaper production facility has one production and packaging line producing 200,000 diapers per day with 95% efficiency.

| | Alt. | Description | Costs | Revenue |
|---|---|---|---|---|
| Mutually Exclusive Revenue alternatives | A | A new line with 240,000 throughput and 98% eff | 6 millions USD | 5,000 USD per day |
| | B | A used line with 200,000 throughput and 96% eff | 2.5 millions USD | 2,000 USD per day |
| | C | A *co-developed* line with 240,000 throughput and 97% eff | 5 millions USD | 3,000 USD per day |
| Independent Projects | 1 | Install an online inspection system. | 100,000 Initial + 2,000 per year | 15,000 USD per year |
| | 2 | Replace existing guiding systems with more accurate ones. | ± 16,000 Initial | 5,000 USD per year |
| | 3 | Buy a new Cutting die made of tungsten carbide material. | 300,000 USD | 40,000 USD per year |

The Do-Nothing alternative is also a valid alternative.

# Present Worth Analysis

Comparing different alternatives to choose one or more, or Nothing



# The Shift: From Math to Engineering

Chapter 2-4 (Before)

$(P/A, i\%, (F/P, i\%, n)$

$(P/F, i\%, n)$

Mathematics

- Goal: Translating cashflows through time
- Abstract Cash Flows
- An interest rate




Chapter 5+ (Now & Later)

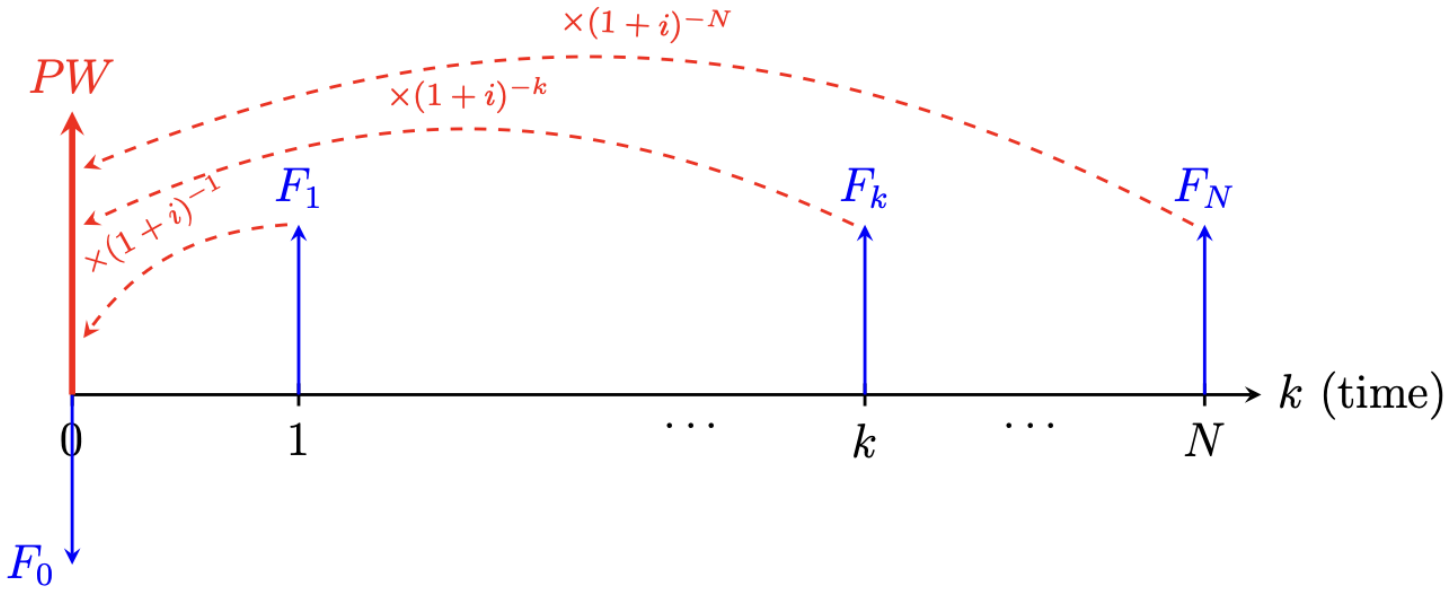


Real-world Projects

- Goal: Evaluating real-world project → Decisions
- Realistic Cash Flows
- Not any interest rate

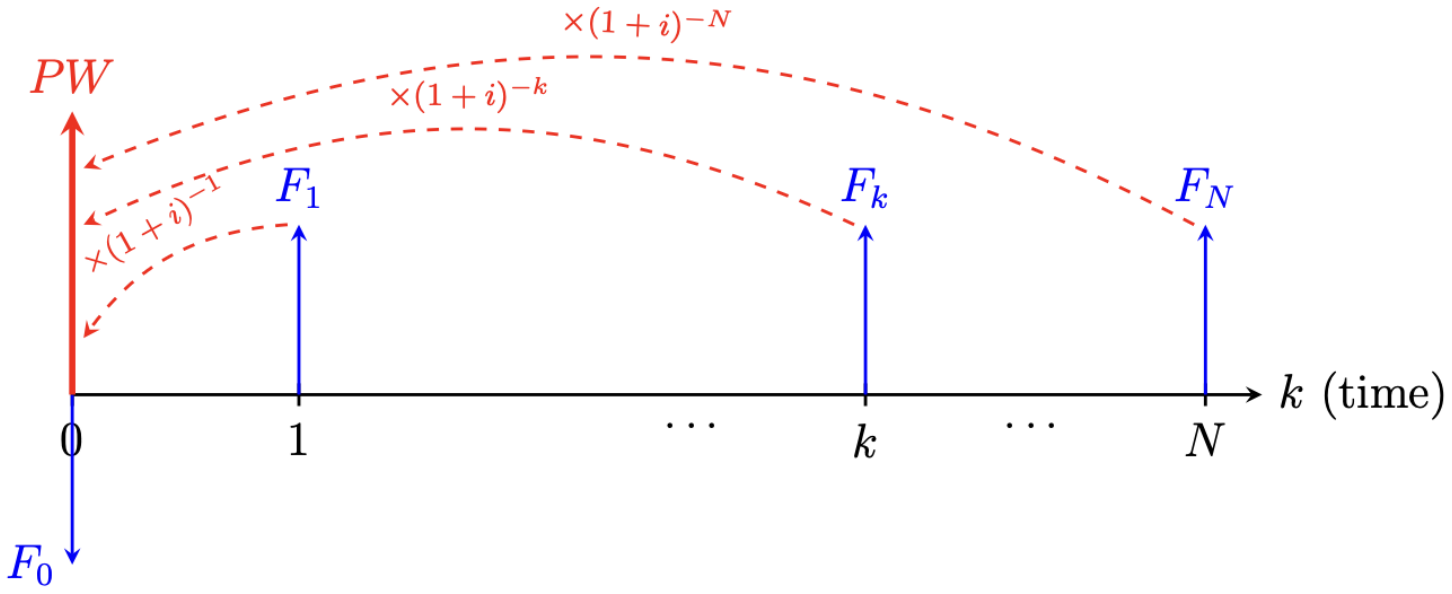
# Present Worth Analysis

Based on the concept of equivalent worth of all cash flows relative to a base or beginning point in time called the present.



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Based on the concept of equivalent worth of all cash flows relative to a base or beginning point in time called the present.




Why MARR?

The return your money would earn if you simply invested it elsewhere (Bonds, stocks, investment portfolio,...)

# PW Analysis – Example (MARR=10%)

| Alt. | Description | Costs | Revenue |
|---|---|---|---|
| A | A new line with 240,000 throughput and 98% eff | 6 millions USD | 5,000 USD per day |
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| 1 | Install an online inspection system. | 100,000 Initial + 2,000 per year | 15,000 USD per year |
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| 3 | Buy a new Cutting die made of tungsten carbide material. | 300,000 USD | 40,000 USD per year |

# PW Analysis – Example Solution

| Alt. | Costs | Revenue | PW |
|---|---|---|---|
| A | 6 millions USD | 5,000 USD per day | **$ 918,185** |
| B | 2.5 millions USD | 2,000 USD per day | **$ 267,274** |
| C | 5 millions USD | 3,000 USD per day | **$ -849,088** |
| 1 | 100,000 Initial + 2,000 per year | 15,000 USD per year | **$ -50,719** |
| 2 | ± 16,000 Initial | 5,000 USD per year | **$ 2,953** |
| 3 | 300,000 USD | 40,000 USD per year | **$ -148,368** |

# PW Analysis – Example Solution

| Alt. | Costs | Revenue | PW |
|---|---|---|---|
| A | 6 millions USD | 5,000 USD per day | **$ 918,185** |
| B | 2.5 mil | | |
| C | | | |
| 1 | | | |
| 2 | ± 16,00 | | |
| 3 | 300,000 USD | 40,000 USD per year | **$ -148,368** |

**Key Assumption**
Equal-service requirement: all alternatives must be compared over the same period

# Different-life alternatives

PW Analysis



# LCM - Example

| Alt. | Description | Costs | Revenue | Expected Life | Salvage Value |
|---|---|---|---|---|---|
| A | A new line with 240,000 throughput and 98% eff | 6 millions USD | 5,000 USD per day | 6 | $ 500,000 |
| B | A used line with 200,000 throughput and 96% eff | 2.5 millions USD | 2,000 USD per day | 3 | $ 200,000 |
| C | A *co-developed* line with 240,000 throughput and 97% eff | 5 millions USD | 3,000 USD per day | 4 | $ 400,000 |
| ~~1~~ | ~~Install an online inspection system.~~ | ~~100,000 Initial + 2,000 per year~~ | ~~15,000 USD per year~~ | | |
| ~~2~~ | ~~Replace existing guiding systems with more accurate ones.~~ | ~~+ 16,000 Initial~~ | ~~5,000 USD per year~~ | | |
| ~~3~~ | ~~Buy a new Cutting die made of tungsten carbide material.~~ | ~~300,000 USD~~ | ~~40,000 USD per year~~ | | |

# Different-Life Alternatives

We must transform into same number of years:

**LCM**: Lowest Common Multiple of lives

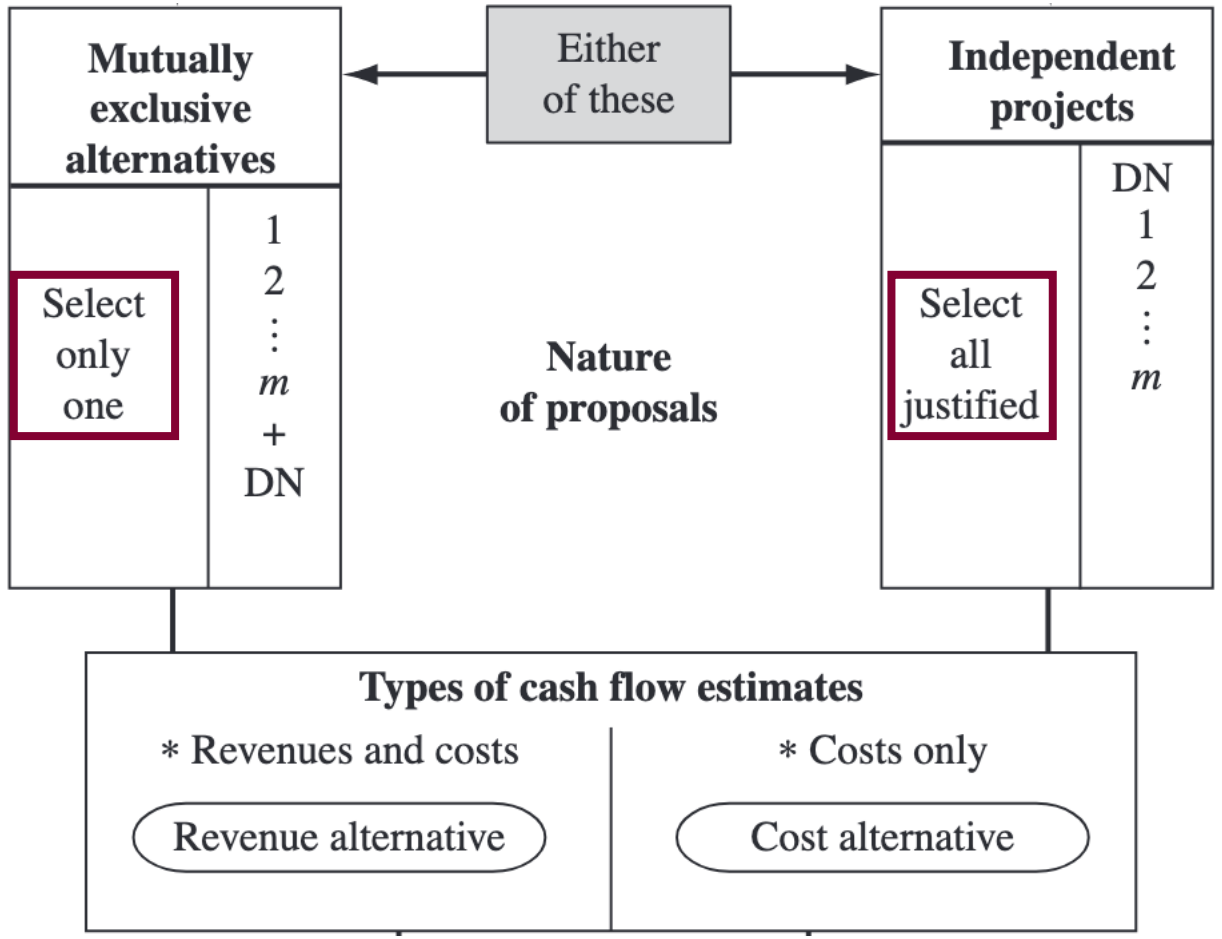
- Assume reinvestment at **same** cash flow estimates in each life cycle of the LCM planning horizon.
- lives of 4 and 6, use $n = 12$

**Study Period**: Assume a fixed planning horizon

- Assume a fixed planning horizon and evaluate the alternatives over it.
- Make some assumption at the end of the horizon depending on alt. life

# Different-Life Alternatives

Select Highest PW




Select all positive PW

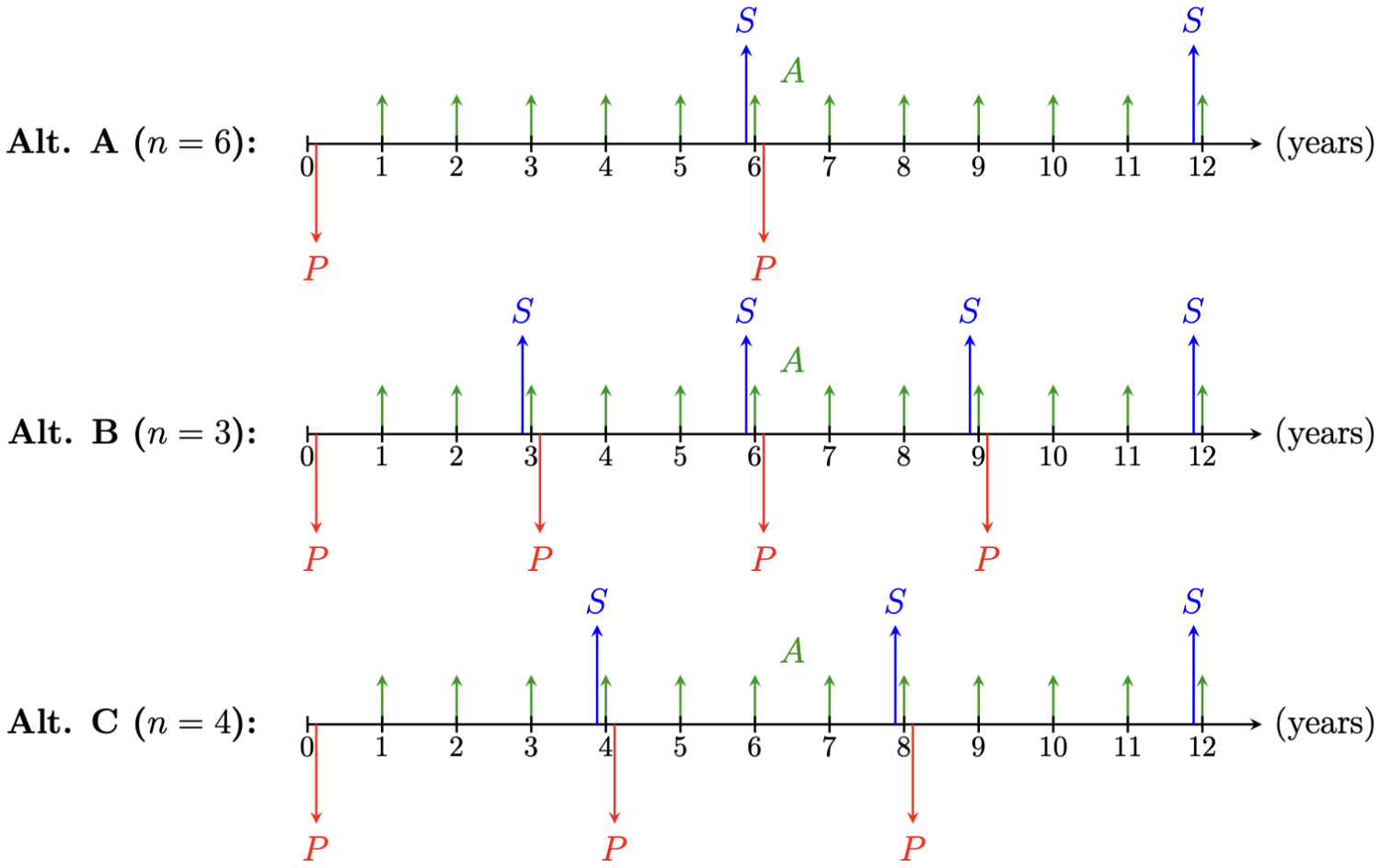
We do not compare independent projects; hence, no need for equal-service assumptions

# LCM - Example

- Find the LCM
- Draw the new extended cash flows for each alternative
- Find the PW of each alternative
- Compare and select



# LCM – Example (Solution)




$$PW_A = -6,000,000 - 6,000,000(P/F, 10\%, 6) + 500,000(P/F, 10\%, 6) + 500,000(P/F, 10\%, 12) + 1,825,000(P/A, 10\%, 12) = \$3,489,696.37$$

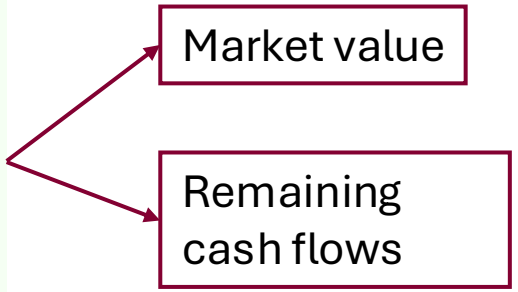
$$PW_B = -2,500,000 - 2,500,000(P/F, 10\%, 3) - 2,500,000(P/F, 10\%, 6) - 2,500,000(P/F, 10\%, 9) + 200,000(P/F, 10\%, 3) + 200,000(P/F, 10\%, 6) + 200,000(P/F, 10\%, 9) + 200,000(P/F, 10\%, 12) + 730,000(P/A, 10\%, 12) = -\$1,464,017.41$$

$$PW_C = -5,000,000 - 5,000,000(P/F, 10\%, 4) - 5,000,000(P/F, 10\%, 8) + 400,000(P/F, 10\%, 4) + 400,000(P/F, 10\%, 8) + 400,000(P/F, 10\%, 12) + 1,095,000(P/A, 10\%, 12) = -\$2,699,350.97$$

# Study Period – 3 scenarios

The table below summarizes the required adjustment for each case.

| Case | Required Adjustment |
|---|---|
| Alt. Life = Study Period | No adjustment required. |
| Alt. Life > Study Period | Estimate a terminal market (or book) value at the end of the study period and treat it as a cash inflow. |
| Alt. Life < Study Period | Compound all net cash flows at the end of the alternative's life forward to the end of the study period at the MARR. |



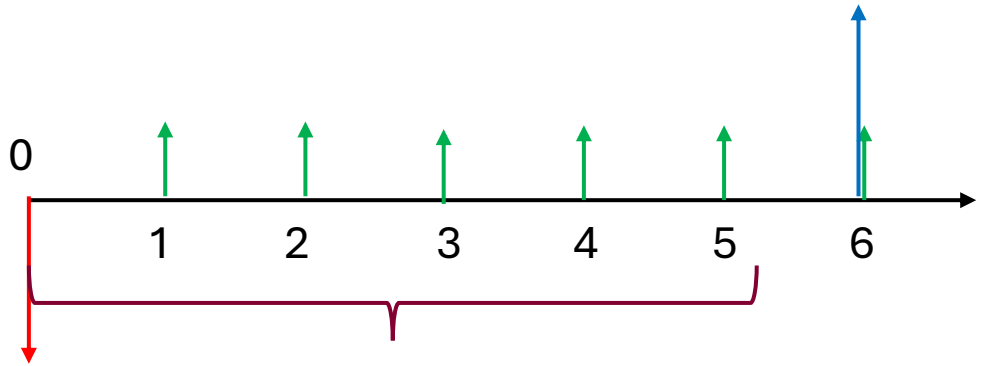
# Study Period - Example

| Alt. | Description | Costs | Revenue | Expected Life | Salvage Value |
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Assume a study period = 5 years

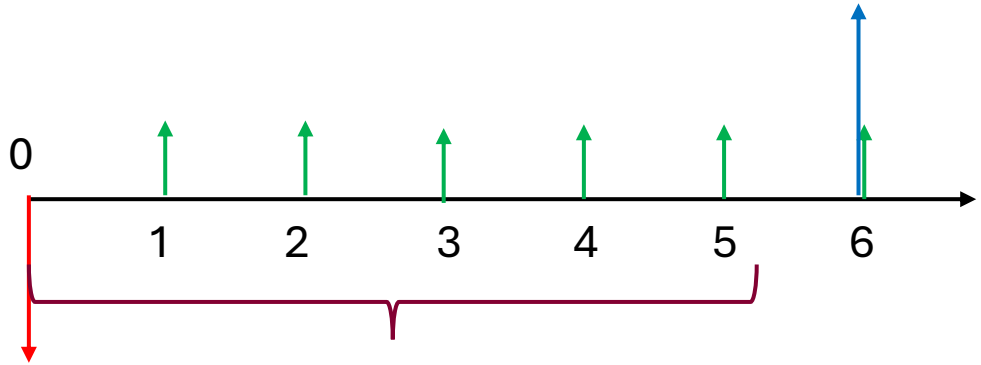
# Study Period – Longer than study period

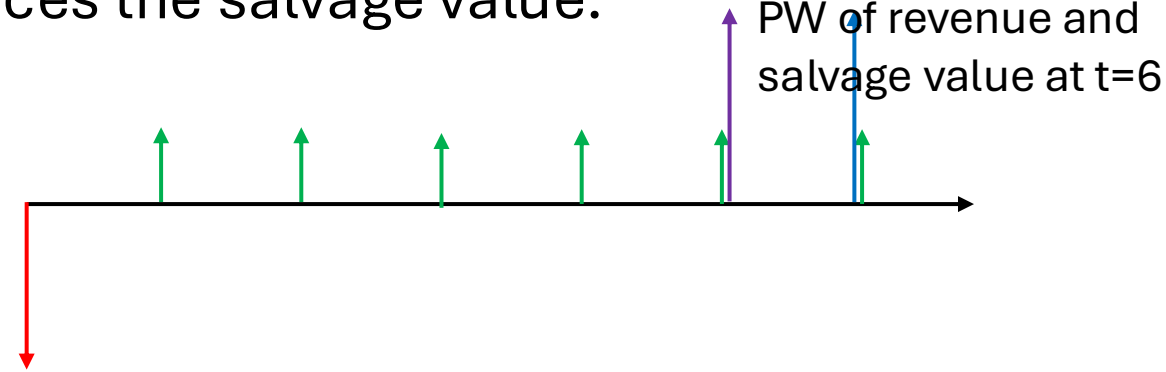
- Alternative A (Longer than study period):
  - Truncate it to 5 years
    - Find the PW of the remaining cash flow
    - If given, use the estimated market value at the 5$^{th}$ year → $ 1,000,000. This estimated market value replaces the salvage value.



# Study Period – Longer than study period

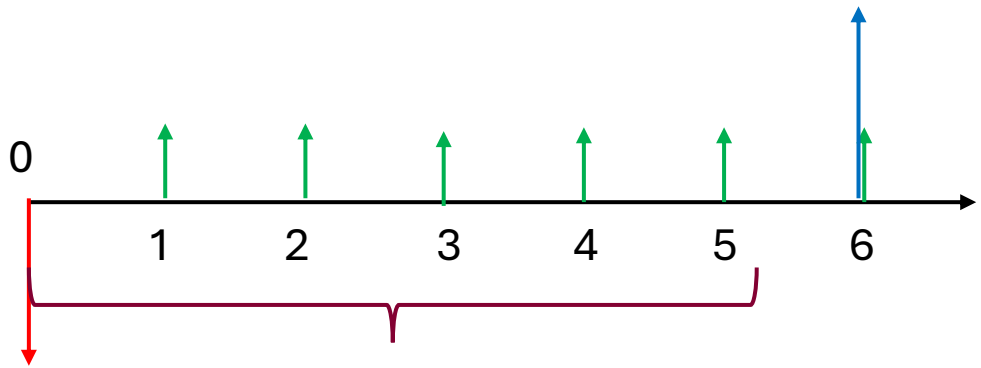
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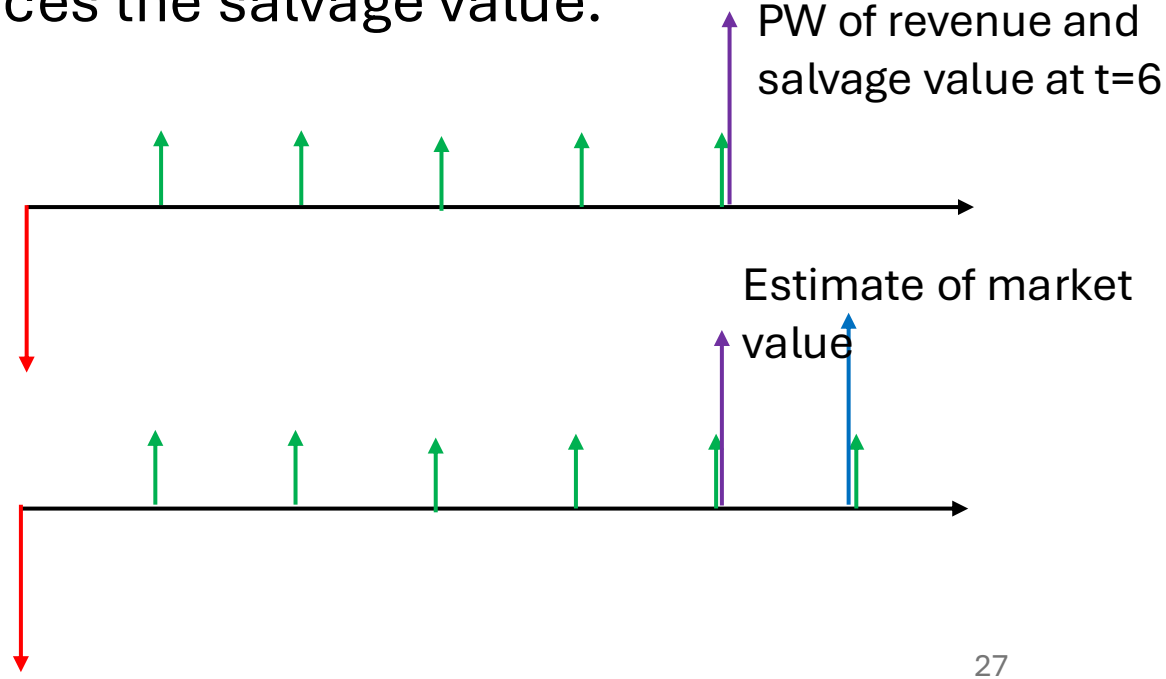


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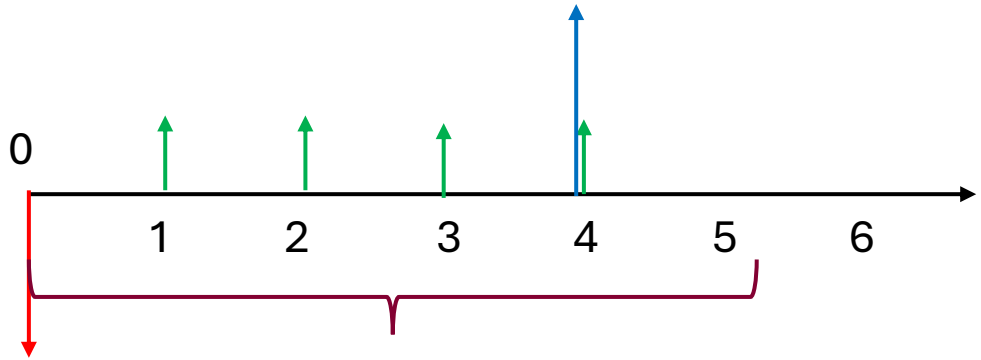
- Alternative A (Longer than study period):
  - Truncate it to 5 years
    - Find the PW of the remaining cash flow
    - If given, use the estimated market value at the 5$^{th}$ year → $ 1,000,000. This estimated market value replaces the salvage value.



# Study Period – Shorter than study period

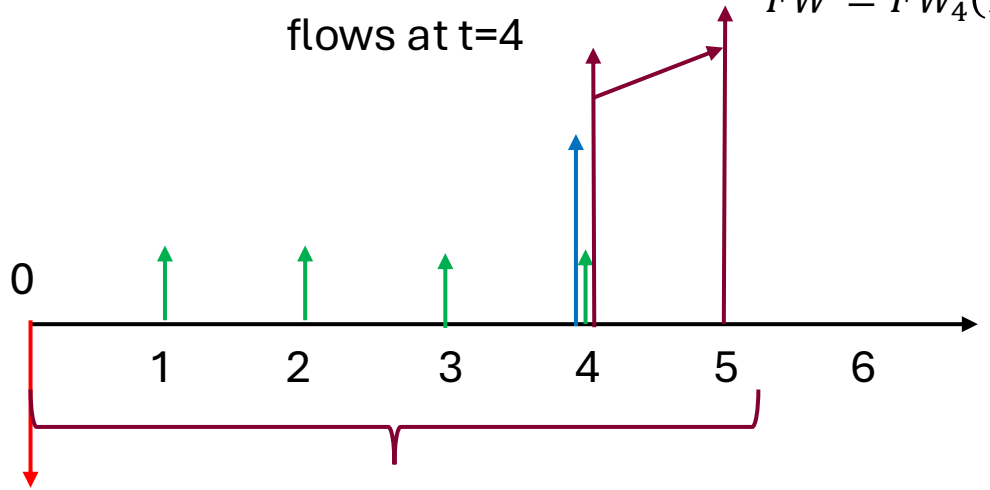
- Alternatives B & C (Shorter than study period):
  - Find the Future Worth of the cash flows at the end of the life.
  - Find the equivalent FW at the end of the study period using the MARR.
  - Compute the PW




FW of cash flows at t=4

FW of cash flows at t=5

$FW = FW_4(F/P, MARR, 1)$



# Other criteria

Another perspectives – not always same results

# Other Criteria

- Future Worth Analysis (FW)
  - Future instead of present
- Capitalized Cost
  - For very long projects
- Payback Period Analysis
  - Not always same results



# Future Worth Analysis

- Similar concept, preferred when:
  - Maximizing future wealth
  - Asset may be sold after some time
  - Project will not “come online” until end of investment period

- FW and PW are equivalent when comparing alternatives



# Capitalized Cost Analysis

- It is the Present Worth
- Alternatives with very long (or infinite) expected useful lives
  - Infrastructure projects (bridges, highways, …),
  - endowments for charity



# CC Analysis - Procedure

1. Cash flow diagram over at least two cycles of periodic cash flows
2. One time cash flows → get the PW
3. Recurring cash flows with a period of $n_R$ (e.g., $R$ \$ every $n_R$ years):
    1. Find equivalent uniform annual worth:

$$A_R = R(^{A}/_{F}, i, n_R)$$

    2. Find equivalent CC for the $A_R$ series:

$$CC_R = \frac{A_R}{i}$$



# CC Analysis - Example

AUB has decided to install a new campus-wide software system called Woohoo. The system has an expected infinite life. The cost estimates are:

- Initial cost: $150,000
- One-time hardware upgrade cost after 10 years: $50,000
- Annual maintenance contract: $5,000 for the first 4 years, and $8,000 thereafter (years 5 to ∞)
- Recurring major software upgrade cost: $15,000 every 13 years

Consider $i = 5\%$



# CC Analysis – Finite and infinite comparison

- Using the study period approach

1. Find the equivalent uniform annual worth ($A$) for one life cycle
2. Compute the CC:

$$CC_{finite} = \frac{A}{i}$$



# Payback Period Analysis

- Estimated time to recover initial investment + a rate of return $i$
- Also based on PW technique
- Find $n_P$

$$0 = -P + \sum_{t=1}^{n_P} NCF_t(^{P}/_{F}, i, t)$$

- Solved by trial-and-error / numerical solvers (Excel)



# Payback Period Analysis - CAUTION

- Not used to make decisions:
  - Disregards subsequent cashflow after the payback period (e.g., salvage value)
  - If $i = 0\%$, fail to account for the time value of money



# Life-cycle Cost

Cost of engineering projects over their complete life cycle



# Phases of an Engineering Project

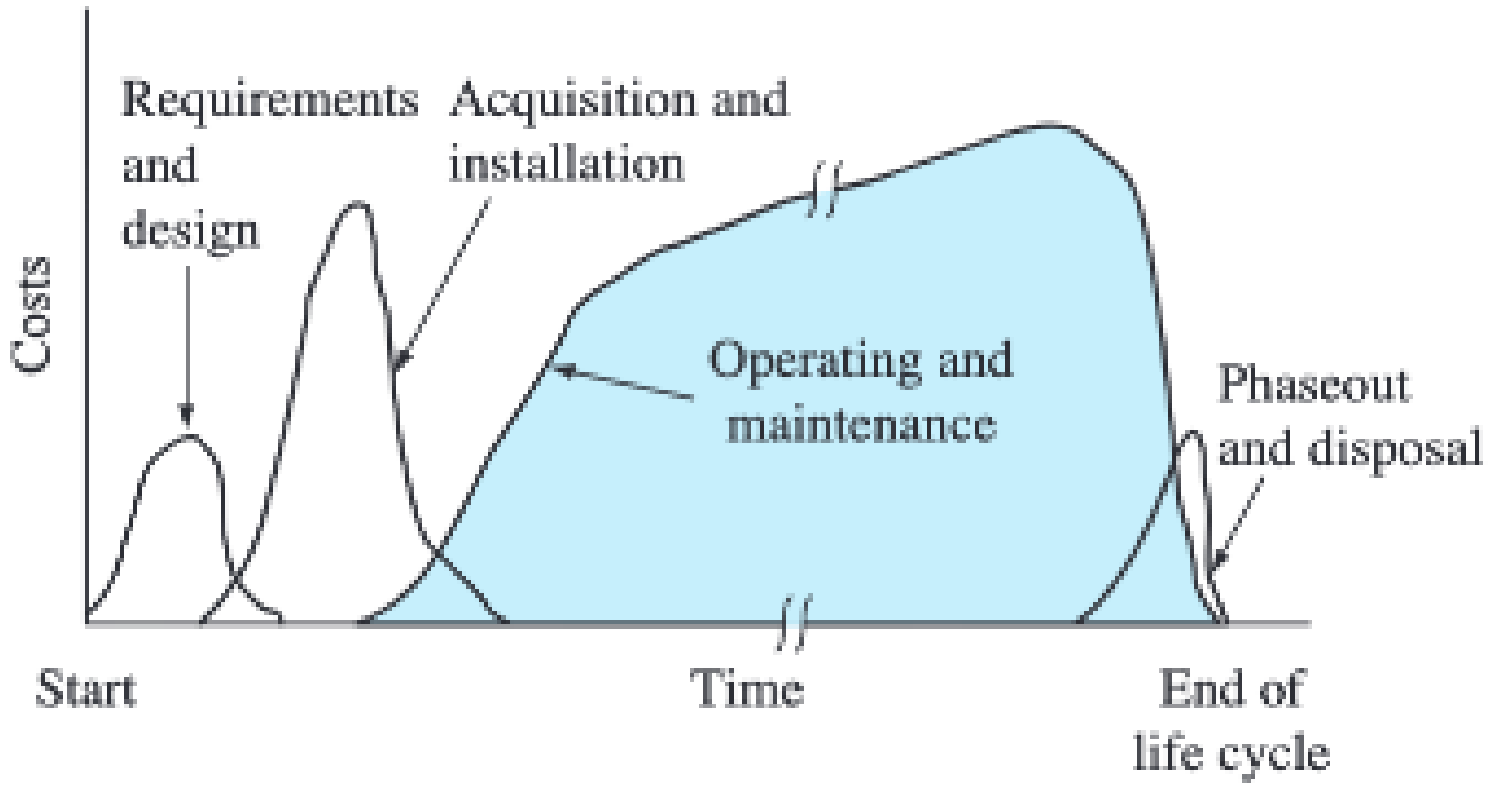



- Requirement definition
- Preliminary design
- Detailed design

- Construction
- Implementation
- Usage

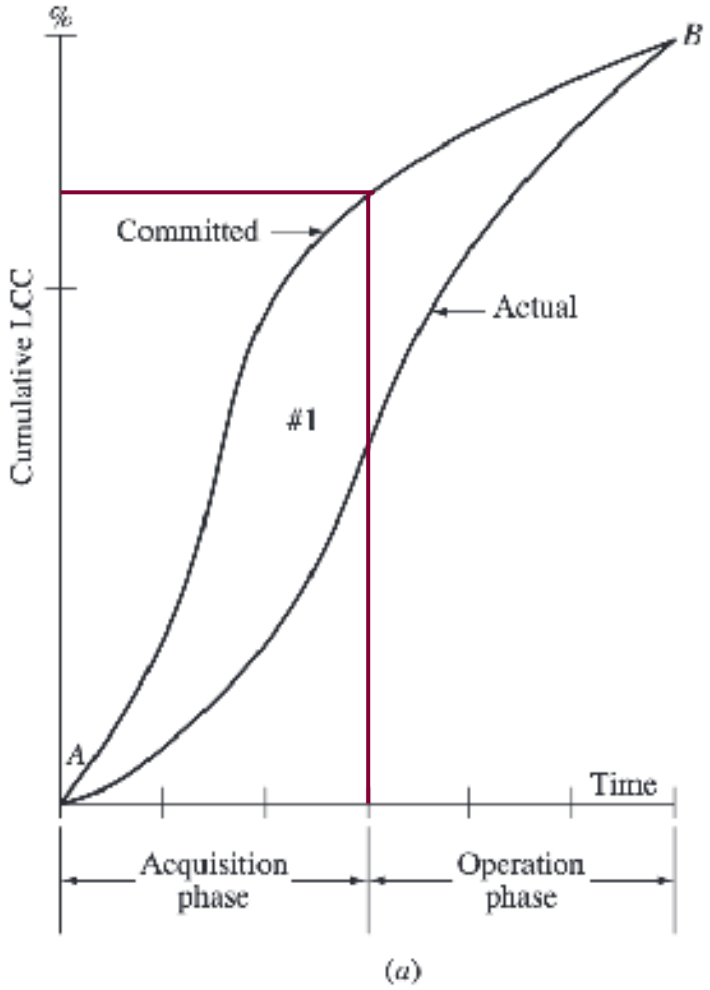
- Recycling
- Removing
- Transitioning

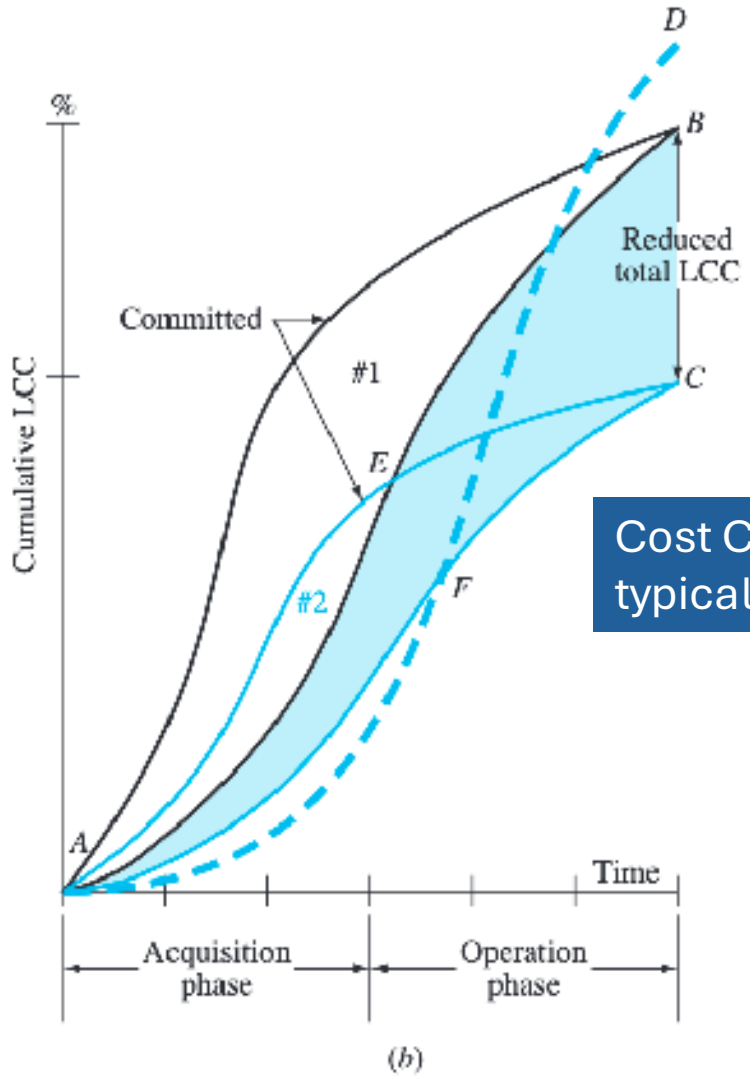
# Phases of an Engineering Project



# LCC – Committed vs Actual costs

75-85%




Improved design reduces LCC

Cost Cutting measures typically backfire